

Machine learning for vision

Winter 2016

Roland Memisevic

Lecture 6, February 16, 2016

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Sampling a continuous signal

The Fourier transform of the Dirac comb with spacing D is the Dirac comb with spacing $\frac{1}{D}$.

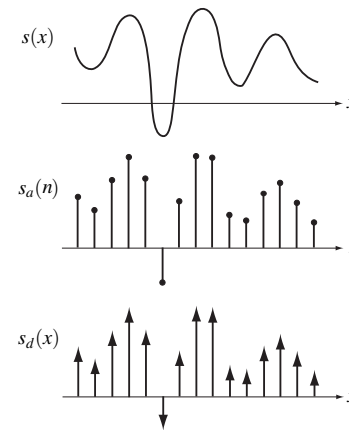
- ▶ Recall that elementwise multiplication in the time domain is convolution in the frequency domain.
- ▶ Sampling therefore generates replicates of the original spectrum known as **(spectral) alias**:

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Sampling a continuous signal



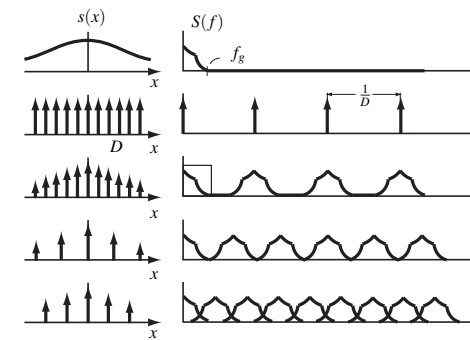
- ▶ Sampling (= A/D conversion) can be formalized as multiplication of a continuous signal by a grid of Dirac deltas = “Dirac comb”.

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Sampling generates spectral aliases



- ▶ The spectrum of the digitized signal is the infinite superposition of translated copies of the original spectrum.

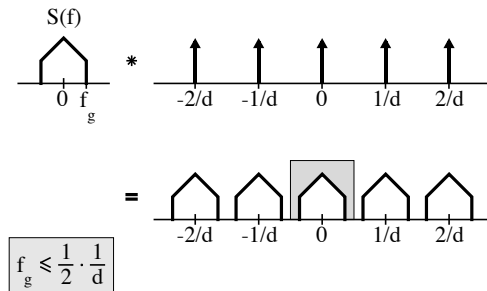
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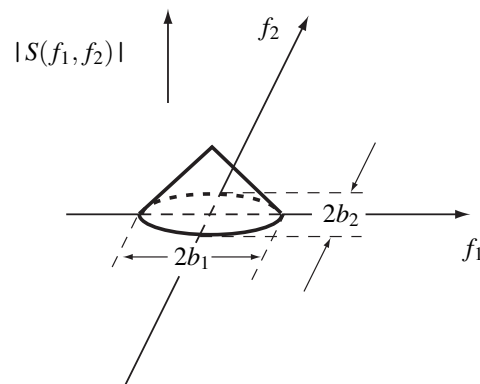
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Sampling generates spectral aliases

- ▶ In theory, we can perfectly undo this effect and recover the original signal by low-pass filtering, if the spectral aliases don't overlap.
- ▶ Otherwise, sampling will lose information.



Sampling theorem in 2d

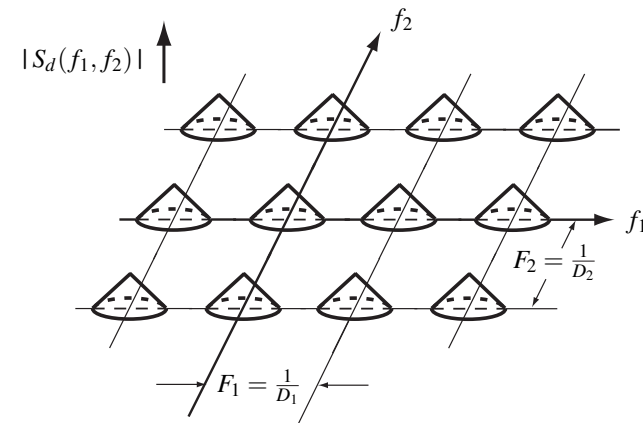


Sampling theorem and Nyquist frequency

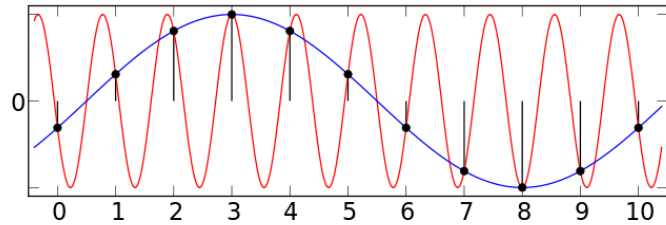
The sampling theorem (wikipedia)

If a function $s(t)$ contains no frequencies higher than D hertz, it is completely determined by giving its ordinates at a series of points spaced $1/(2D)$ seconds apart. The frequency $1/(2D)$ is called Nyquist frequency.

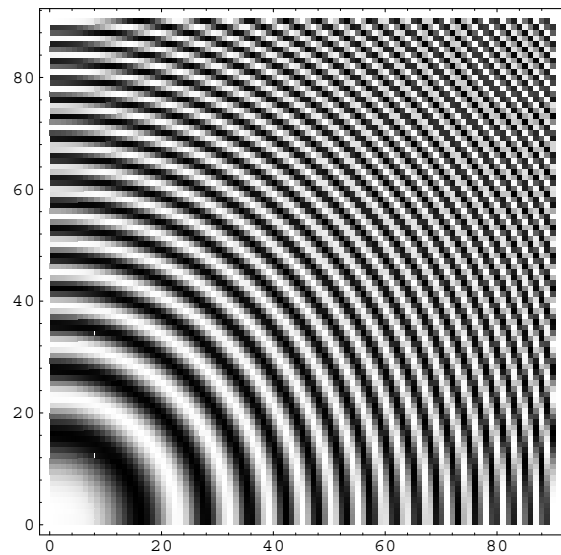
Sampling theorem in 2d



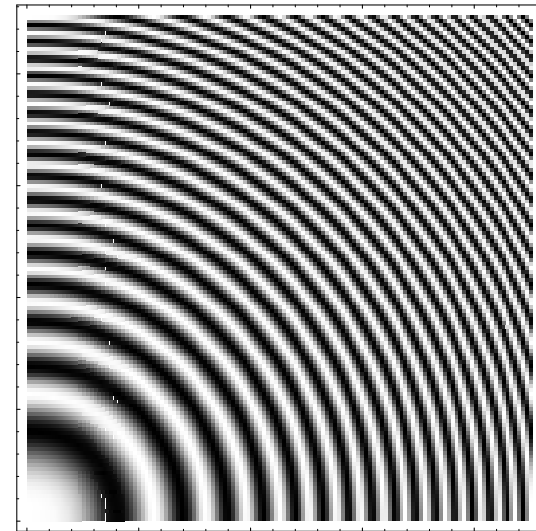
Aliasing example



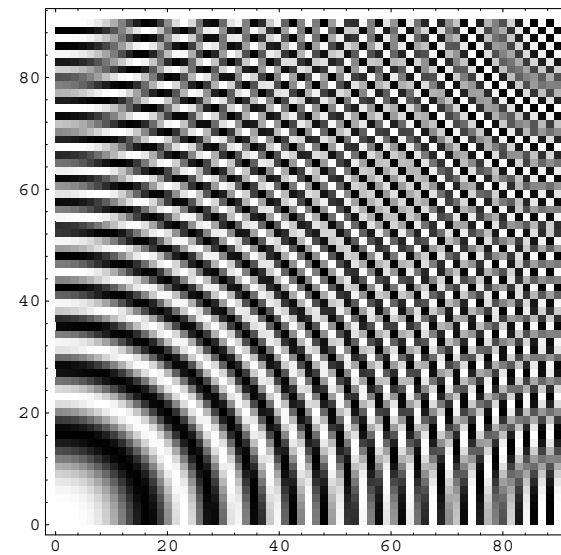
Aliasing in 2d



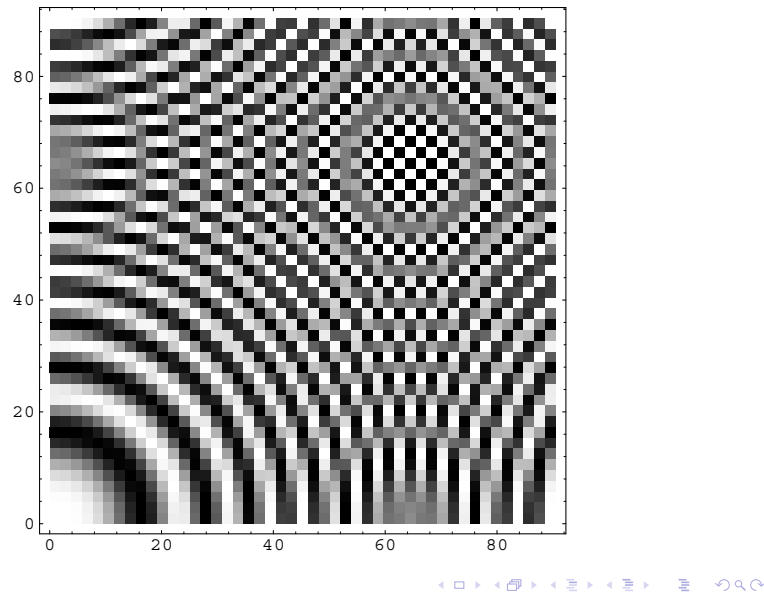
Aliasing in 2d



Aliasing in 2d



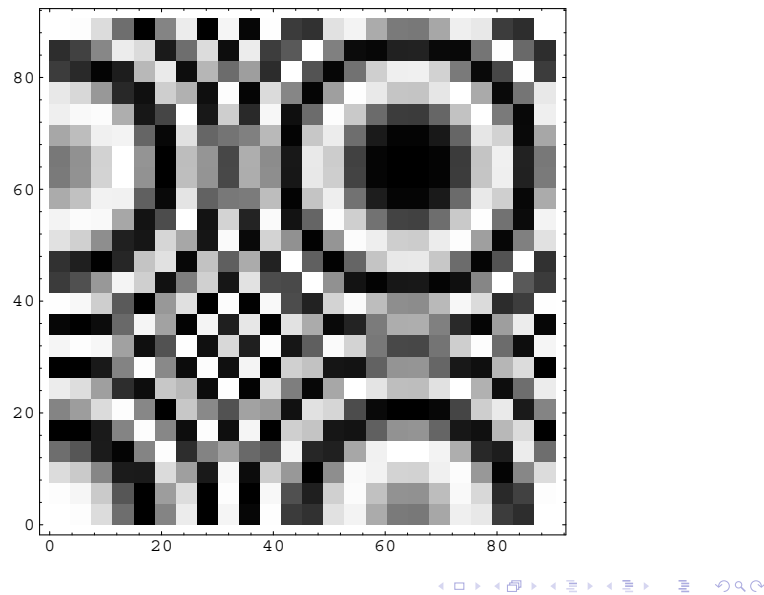
Aliasing in 2d



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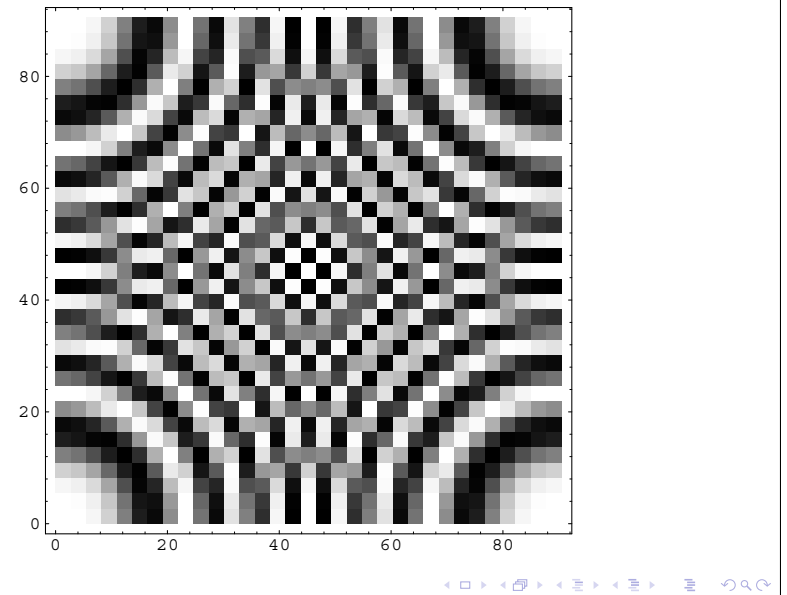
Aliasing in 2d



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Aliasing in 2d



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Aliasing in 2d



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Aliasing in 2d



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Smoothing

- An immediate practical implication of the sampling theorem:

If you want to subsample an image, do smoothing first.

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Aliasing in 2d



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Example

Welcome to Joe's webpage! $\xrightarrow{\text{subsampling}}$ Welcome to Joe's webpage!

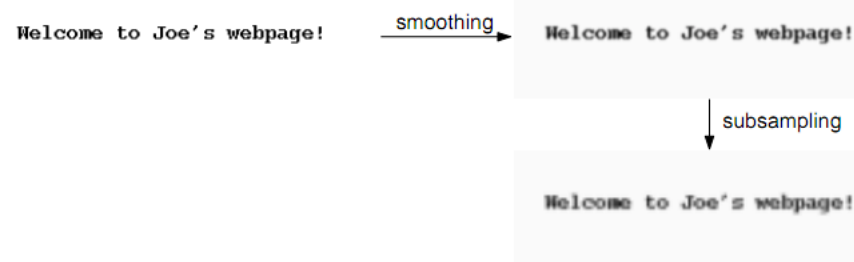
example from
<http://redwood.berkeley.edu/bruno/psc129/handouts/aliasing.pdf>

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Example



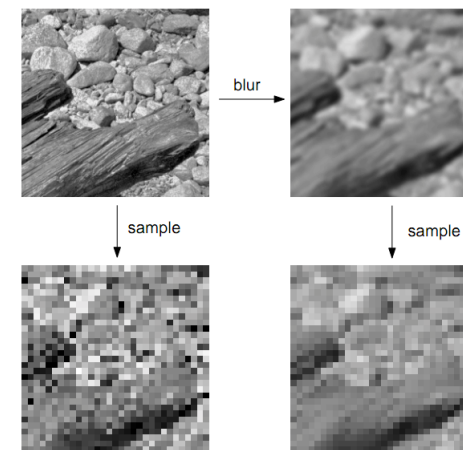
example from
<http://redwood.berkeley.edu/bruno/psc129/handouts/aliasing.pdf>

Ideal interpolation

$$\begin{aligned}
 & \begin{array}{c} S(f) \\ \text{---} \text{---} \text{---} \text{---} \text{---} \\ | \quad | \quad | \quad | \quad | \\ 0 \quad f \quad g \end{array} * \begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \\ -2/d \quad -1/d \quad 0 \quad 1/d \quad 2/d \end{array} \\
 &= \begin{array}{c} \text{---} \text{---} \text{---} \text{---} \text{---} \\ | \quad | \quad | \quad | \quad | \\ -2/d \quad -1/d \quad 0 \quad 1/d \quad 2/d \end{array} \\
 & \boxed{f_g \leq \frac{1}{2} \cdot \frac{1}{d}}
 \end{aligned}$$

- ▶ To re-generate the original (continuous) signal from the sampled signal, do low-pass filtering.
- ▶ Applying an ideal low-pass filter amounts to convolving the signal with the Fourier transform of the box function.
- ▶ Thus, **ideal interpolation is convolution of the discrete signal with the sinc-function.**

Example

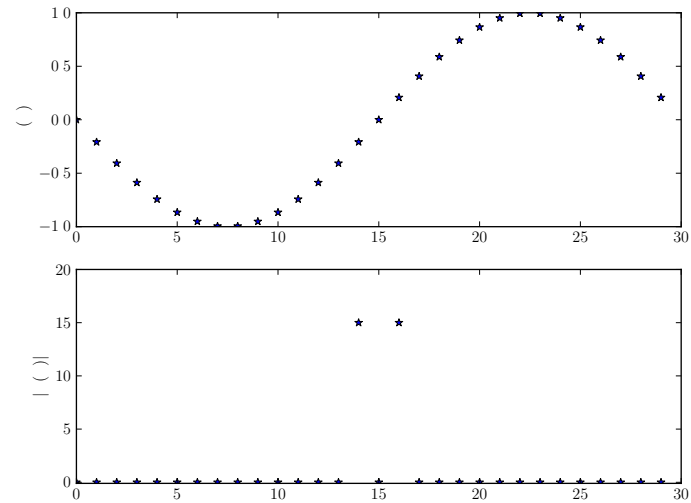


example from
<http://redwood.berkeley.edu/bruno/psc129/handouts/aliasing.pdf>

DFT leakage

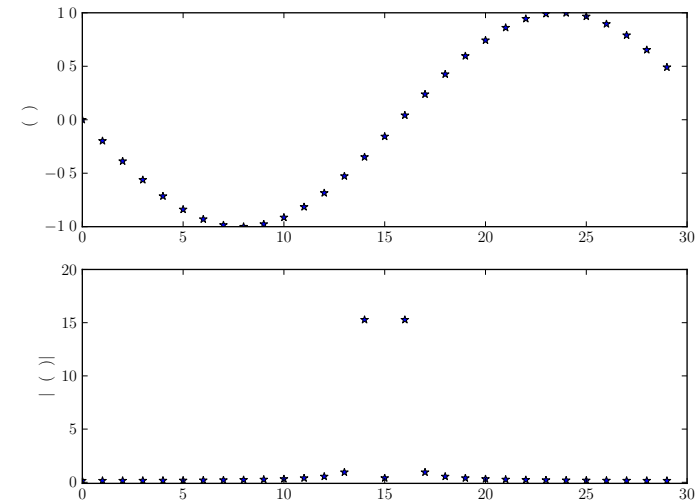
- ▶ We can think of the DFT of a finite signal as the DFT of a periodic signal after multiplying it by a rectangular window.
- ▶ The DFT spectrum you get can be thought of as the spectrum of the periodic signal convolved with a sinc-function.
- ▶ Because of the zero-crossings of the sinc-function the convolution will have no effect on signal components whose frequencies are integer multiples of the window length.
- ▶ For any other components, the convolution will generate additional components in the spectrum.
- ▶ This effect is known as **leakage**.

Leakage example



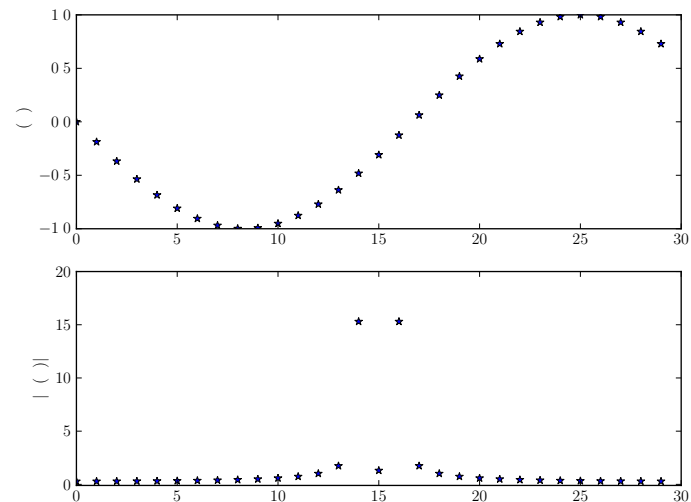
Navigation icons: back, forward, search, etc.

Leakage example



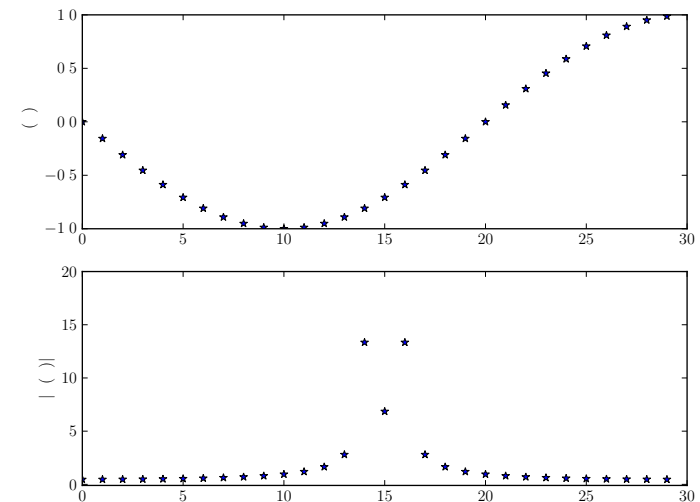
Navigation icons: back, forward, search, etc.

Leakage example



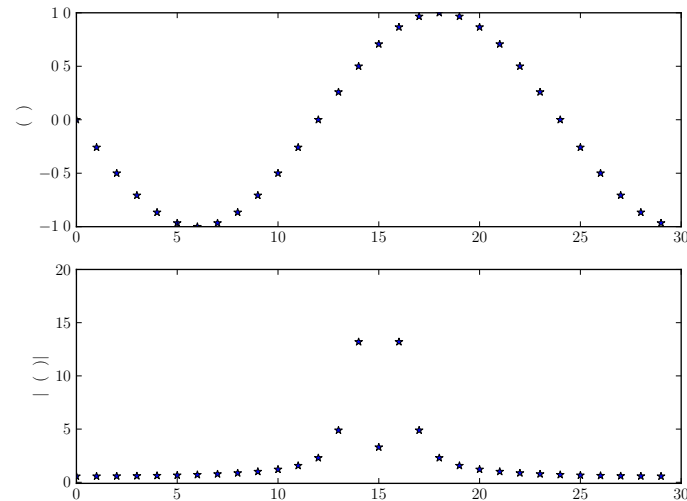
Navigation icons: back, forward, search, etc.

Leakage example



Navigation icons: back, forward, search, etc.

Leakage example



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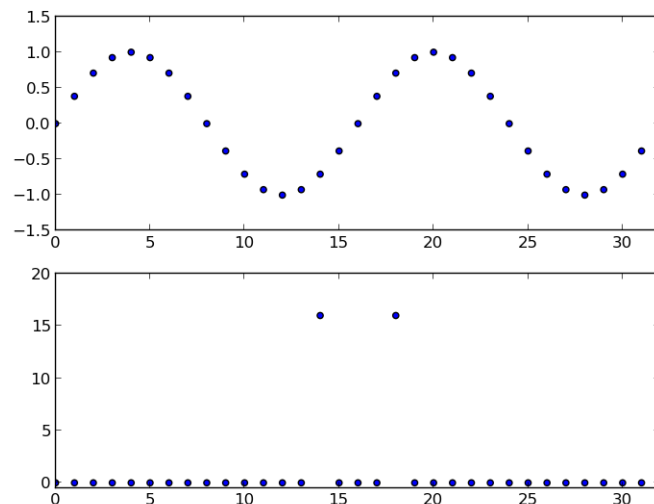
Windowing

- ▶ It is best to think of leakage as *leaking into*: frequency components or noise can leak into the frequency bins that you actually want to detect.
- ▶ Leakage cannot be avoided.
- ▶ But a window other than the box-window may lead to different, potentially less undesirable, leakage properties.
- ▶ This may allow us to re-distribute the effect of leakage across the bins such that it causes the least harm for the application at hand.

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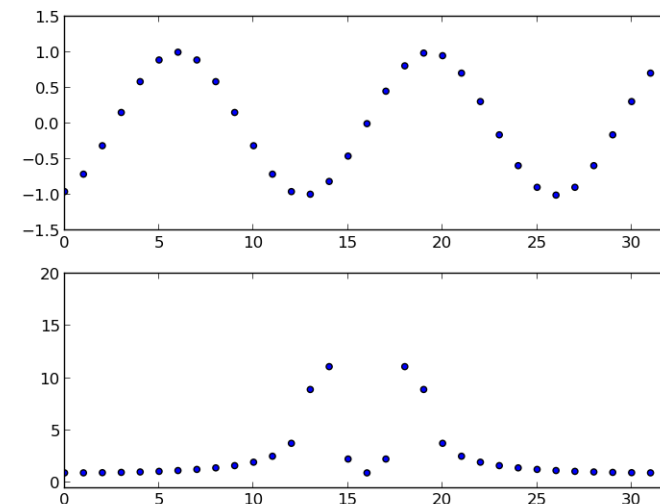
Leakage with box window



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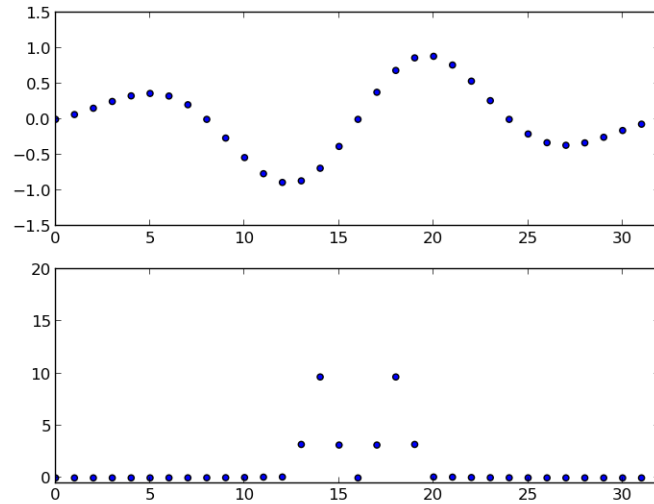
Leakage with box window



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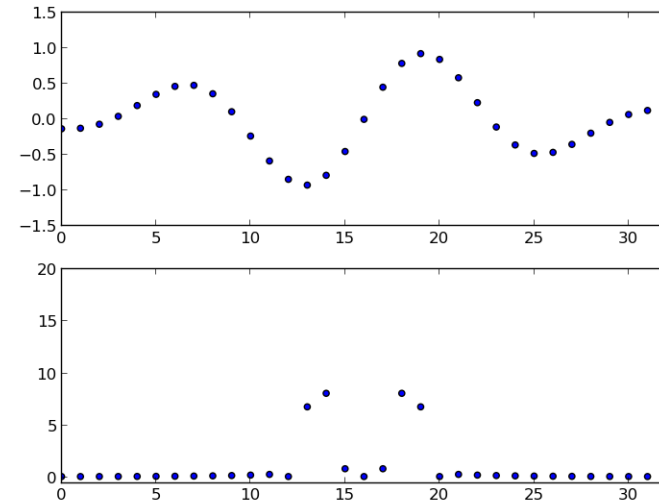
Leakage with Gaussian window



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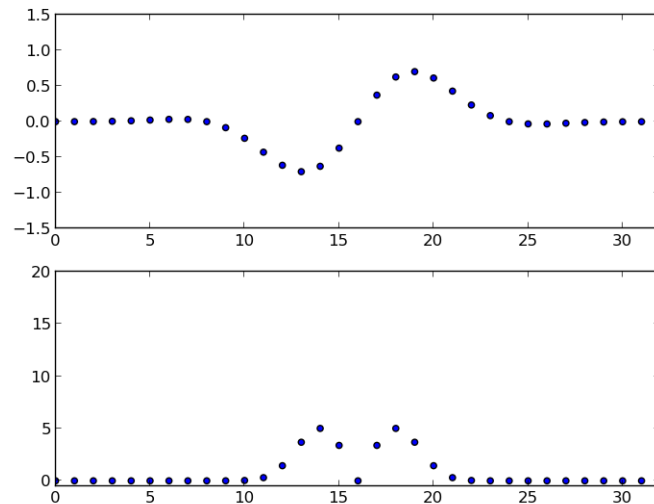
Leakage with Gaussian window



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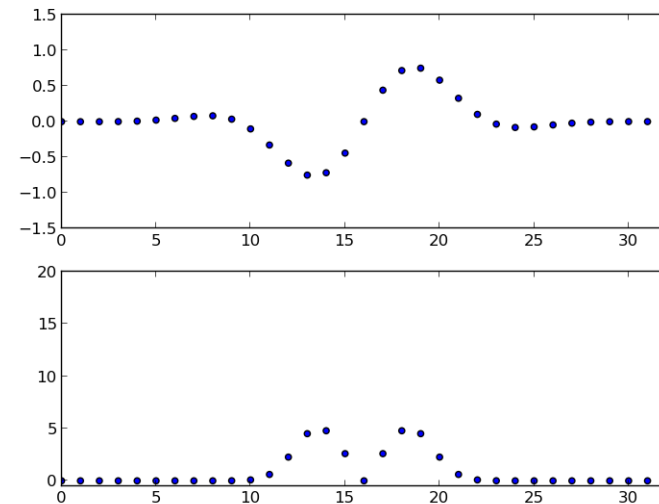
Leakage with small Gaussian window



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Leakage with small Gaussian window



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Windowing and Short Time Fourier Transform

- ▶ The **Short-Time Fourier Transform (STFT)** is an application of window functions:
- ▶ Fourier-transform the signal *locally*, then view the resulting set of spectra as a function of time or space.
- ▶ In 1d, the result is called *spectrogram*.
- ▶ The choice of window is important for the reasons just discussed.
- ▶ An STFT using a Gaussian window is also called **Gabor transform**.

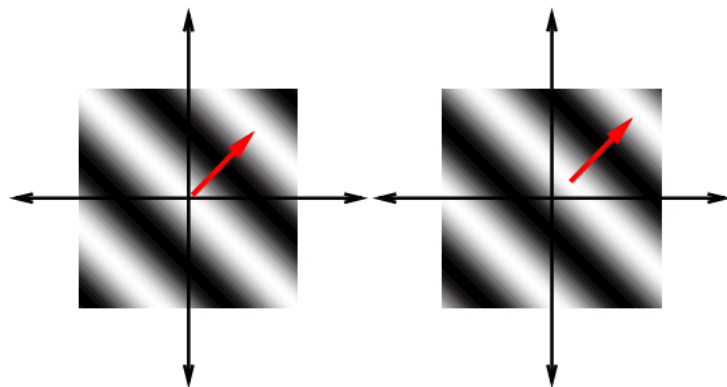
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Example

Wave:



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Gabor features (2d)

- ▶ 2d Gabor functions are highly common in visual recognition tasks.
- ▶ They can be defined as the product of two exponential functions (one for the window, one for the phasor):

$$g(K, \sigma, x_0, y_0, \gamma, u, v, P) = K \exp \left(-\frac{1}{\sigma^2} ((x - x_0)^2 + \gamma^2 (y - y_0)^2) \right) \cdot \exp(i2\pi(ux + vy) + P)$$

- ▶ Many variations are possible (for example, Gabor filters with oriented rather than axis parallel windows, etc.).

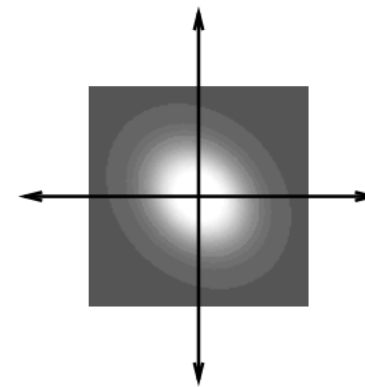
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Example

Window:



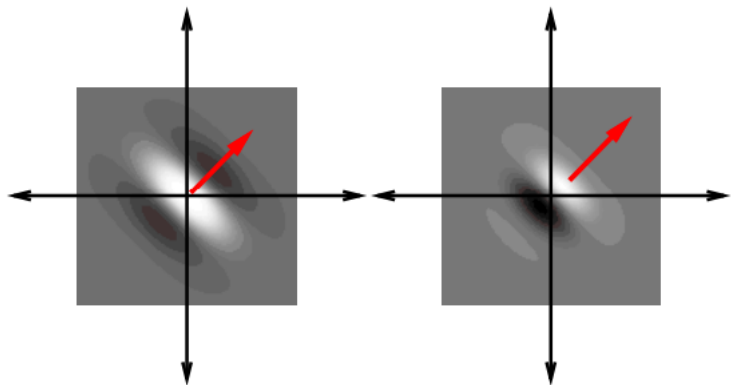
A set of navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.

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Example

Gabor feature:



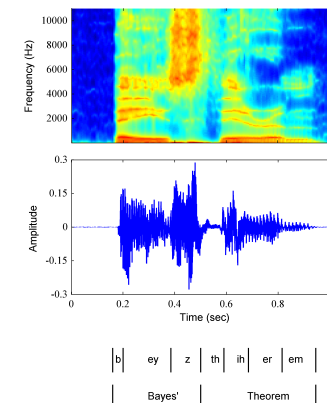
Frequency channels

- ▶ In many applications, local Gabor features are used as *filters*, ie. they are scanned across the image.
- ▶ This naturally raises the question:
- ▶ What is the frequency response of a Gabor filter?
- ▶ It is a *localized blob* in the frequency domain:
- ▶ The Fourier transform of a phasor times a Gaussian will be a delta-peak convolved with a Gaussian.
- ▶ Thus, Gabor filters are **oriented bandpass filters**.

Frequency channels

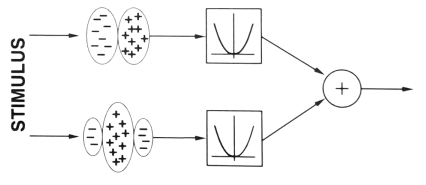
- ▶ In many applications, local Gabor features are used as *filters*, ie. they are scanned across the image.
- ▶ This naturally raises the question:
- ▶ What is the frequency response of a Gabor filter?

A spectrogram (top) of an utterance



- ▶ (from Bishop, 2006)
- ▶ The visual analog of the spectrogram is a 3-dimensional feature map.

Complex cells



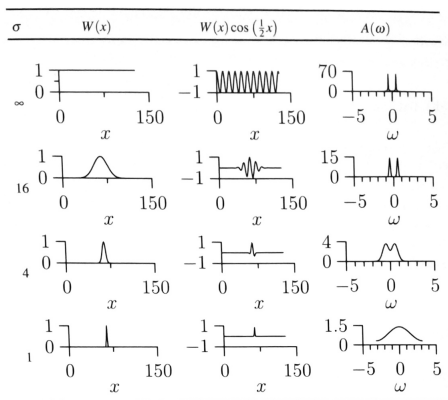
- ▶ AKA “energy mechanism”
- ▶ A Fourier feature pair with 90 deg phase difference is known as **quadrature pair**. (Better than saying “sin/cos”, because phasors in the Fourier expansion may have arbitrary different phase.)

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The uncertainty principle



from: Natural Image Statistics (Hyvarinen, Hurri, Hoyer; 2009)

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Why spatial pooling computes approximate amplitude spectra

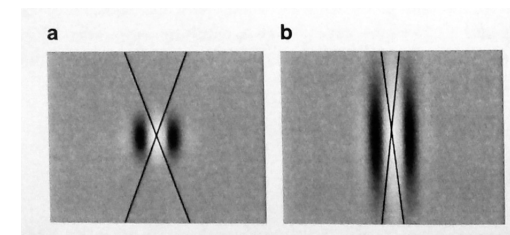
- ▶ In a convnet, lower layers learn Gabor-like features.
- ▶ Thus, locally, a feature map with ReLU units contains the rectified responses of phase-shifted Fourier components.
- ▶ Local max-pooling will keep the absolute value of the best matching phase.
- ▶ Local average-pooling will keep an “attenuated” version of this value.
- ▶ Overall, pooling therefore computes a “sample-approximation” of the amplitude response.

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In 2d: orientation uncertainty



from: Natural Image Statistics (Hyvarinen, Hurri, Hoyer; 2009)

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Some recent papers on Fourier analysis in convnets

- ▶ “Fast Training of Convolutional Networks through FFTs” <http://arxiv.org/pdf/1312.5851v5.pdf>
- ▶ “Spectral Representations for Convolutional Neural Networks” <http://papers.nips.cc/paper/5649-spectral-representations-for-convolutional-neural-networks.pdf>
- ▶ “On Random Weights and Unsupervised Feature Learning”
<http://www.robotics.stanford.edu/~ang/papers/nipsdluf10-RandomWeights.pdf>